

The Chiral Dirac Equation Challenges: Hermiticity, Neutrino Mass Formalism, and Extensions to Higher Spin Fields.

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Abstract

The Chiral Symmetric Dirac Equation (DECS), derived from the irreducible representations of the Poincaré group, introduces an essential chiral angle (α) and an associated pseudoscalar mass term. This work addresses critical theoretical challenges arising from this generalization of the Dirac equation. We begin by formally establishing the physical Hermiticity of the DECS Hamiltonian by demonstrating that the complex mass term $m e^{-i2\alpha\gamma^5}$ can be reduced to a purely scalar mass through a global chiral field rotation, which ensures real eigenvalues for the system's energy. Subsequently, we resolve the inherent contradiction of deriving Dirac mass for the left-chiral neutrino field (νL) in the absence of a right-chiral component (νR). We propose a coherent mass mechanism rooted in the $\Delta L = 2$ violation of the Majorana formalism, providing a physically consistent theoretical framework for neutrino mass generation within the DECS context. Finally, we explore the conceptual role of the chiral degree of freedom in relativistic wave equations for fields with higher spin (e.g., Proca and Rarita-Schwinger), thereby laying the groundwork for future extensions of the chiral formalism beyond spin 1/2. Our findings affirm the consistency of the Chiral Dirac Equation and contribute to a more general understanding of chiral structure in Quantum Field Theory.

Keywords: Chiral Symmetry, Dirac Equation, Majorana Mass, Neutrino Mass, Hermiticity, Poincaré Group, Clifford Algebra, Pseudoscalar Mass, Higher Spin Fields, Yukawa Coupling, Spin 1/2.

1 Introduction

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2 Methods

We adopt the formalism of relativistic quantum field theory on a flat, non-interacting Minkowski spacetime. The background geometry is defined by the orthonormal basis $\{\gamma^\mu\}$ satisfying the Clifford algebra $\mathcal{Cl}_{1,3}(\mathbb{C})$, given by the anticommutation relation

$$\{\gamma^\mu, \gamma^\nu\} \equiv \gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2\eta^{\mu\nu}, \quad (1)$$

where $\eta^{\mu\nu}$ is the Minkowski metric. Consequently, $(\gamma^0)^2 = 1$ and $(\gamma^k)^2 = -1$, where Greek indices $\mu = 0, \dots, 3$ denote spacetime coordinates and Latin indices $k = 1, \dots, 3$ denote spatial components.

The chiral structure is determined by the pseudoscalar $\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3$, where i is the imaginary unit. Since the spacetime dimension is even, γ^5 anticommutes with all vector generators γ^μ and squares to the identity. This permits the construction of the orthogonal chiral projectors

$$P_L = \frac{1}{2}(1 - \gamma^5), \quad P_R = \frac{1}{2}(1 + \gamma^5), \quad (2)$$

satisfying $P_L + P_R = 1$ and $P_L P_R = 0$. A spinor field ψ therefore decomposes as $\psi = P_L \psi + P_R \psi \equiv \psi_L + \psi_R$.

2.1 Hamiltonian Formulation of the Chiral Dirac Equation

Following the formulation of Watson and Musielak [1–3], we consider the Chiral Dirac Equation (CDE) for a spinor field ψ :

$$(i\gamma^\mu \partial_\mu - m e^{-i\alpha\gamma^5})\psi = 0. \quad (3)$$

The corresponding effective Lagrangian density is given by

$$\mathcal{L}_{\text{eff}} = \bar{\psi}(i\gamma^\mu \partial_\mu - m e^{-i\alpha\gamma^5})\psi, \quad (4)$$

where $\bar{\psi} = \psi^\dagger \gamma^0$ is the Dirac adjoint. We derive the Hamiltonian density $\mathcal{H}$ via the Legendre transform, using the canonical momentum $\pi = \partial \mathcal{L}_{\text{eff}} / \partial \dot{\psi}$:

$$\mathcal{H} = \pi \dot{\psi} - \mathcal{L}_{\text{eff}}. \quad (5)$$

Explicit calculation yields:

$$\mathcal{H} = \psi^\dagger (-i\gamma^{0k} \partial_k + \gamma^0 m e^{-i\alpha\gamma^5})\psi. \quad (6)$$

3 Hermitiiiiiiiiiii

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4 Chiral Seesaw Mechanism

As an alternative approach to explain the smallness of neutrino masses, we revisit the mass generation mechanism proposed by Watson and Musielak within the framework of the Chiral Dirac Equation (CDE) [1, 2]. In this formalism, the fermion mass arises from Yukawa couplings to both a scalar field ϕ_1 (Standard Model-like) and a pseudoscalar field ϕ_2 .

In Watson's proposal [3], restricting the theory to pure left-handed fermionic fields causes the mass terms to vanish identically. Considering the original Yukawa Lagrangian postulated by Watson:

$$\mathcal{L}_Y = -\frac{\lambda_1 v_1}{\sqrt{2}} \bar{\nu} \nu - \frac{\lambda_2 v_2}{\sqrt{2}} \bar{\nu} \gamma^5 \nu \quad (7)$$

Both terms vanish if we impose the left-handed chirality constraint on the neutrino field ($P_L \nu = \nu$, $P_R \nu = 0$). The scalar product $\bar{\nu} \nu$ vanishes because the Dirac adjoint of a left-handed field behaves as a right-handed projector acting to the left:

$$\begin{aligned} \bar{\nu} &= \overline{P_L \nu} = \nu^\dagger P_L \gamma^0 \\ &= \nu^\dagger \gamma^0 P_R = \bar{\nu} P_R. \end{aligned}$$

Multiplying by ν from the right, and recalling that $\nu = P_L \nu$, we obtain:

$$\bar{\nu} \nu = (\bar{\nu} P_R)(P_L \nu) = \bar{\nu} \underbrace{(P_R P_L)}_0 \nu = 0.$$

Similarly, for the pseudoscalar term $\bar{\nu} \gamma^5 \nu$, we utilize the fact that γ^5 commutes with the chiral projectors ($[\gamma^5, P_{L/R}] = 0$). Following the previous result:

$$\begin{aligned} \bar{\nu} \gamma^5 \nu &= (\bar{\nu} P_R) \gamma^5 (P_L \nu) \\ &= \bar{\nu} \gamma^5 \underbrace{(P_R P_L)}_0 \nu \\ &= 0. \end{aligned}$$

Thus, without the inclusion of a right-handed component, no Dirac mass term can be generated in this framework. However, if we assume the existence of a right-handed neutrino component ν_R (such that $\nu = \nu_L + \nu_R$), the Dirac mass term does not vanish. Instead, it acquires a complex phase determined by the chiral angle α , which is intrinsic to the CDE. To naturally suppress the neutrino mass without relying on fine-tuning, we invoke the Majorana mass term, leading to a Chiral Seesaw Mechanism.

4.1 The Chiral Dirac Mass Term

Following Watson and Musielak [1], the general Yukawa Lagrangian expanded about the vacuum expectation values (VEVs) v_1 and v_2 corresponds to (7). Defining the scalar

mass parameter $M = \lambda_1 v_1 / \sqrt{2}$ and the pseudoscalar mass parameter $\tilde{M} = \lambda_2 v_2 / \sqrt{2}$, and decomposing the fields into their chiral components ($\nu_{L,R} = P_{L,R} \nu$), the Dirac mass term becomes:

$$\mathcal{L}_{\text{Dirac}} = \bar{\nu}_L (M + \tilde{M}) \nu_R + \bar{\nu}_R (M - \tilde{M}) \nu_L. \quad (8)$$

In the CDE formalism, these parameters are related to the propagation mass m and the chiral angle α by $M = m \cos 2\alpha$ and $\tilde{M} = -im \sin 2\alpha$. Consequently, the effective Dirac mass coupling M_D connecting $\bar{\nu}_L$ and ν_R is identified as:

$$M_D \equiv M + \tilde{M} = m(\cos 2\alpha - i \sin 2\alpha) = m e^{-2i\alpha}. \quad (9)$$

The result in (9) is crucial: the Dirac mass in this model carries a physical phase derived from the underlying chiral symmetry breaking.

4.2 The Full Majorana-Dirac Lagrangian

To implement the seesaw mechanism, we introduce Majorana mass terms. Unlike Dirac terms which couple left- and right-handed fields ($\bar{\nu}_L \nu_R$), Majorana mass terms violate lepton number by two units. Specifically, we utilize the charge-conjugated right-handed field $\nu_R^c = C \bar{\nu}_R^T$.

The most general Lorentz-invariant mass Lagrangian including both the Chiral Dirac term derived above and the Majorana terms is:

$$\mathcal{L}_{\text{Mass}} = \bar{\nu}_L M_D \nu_R + \frac{1}{2} \bar{\nu}_L^c M_L \nu_L + \frac{1}{2} \bar{\nu}_R^c M_R \nu_R + \text{h.c.} \quad (10)$$

Here, M_L represents the Majorana mass for the active left-handed neutrinos (assumed to be negligible or zero in the minimal Type-I seesaw), and M_R is the heavy Majorana mass scale for the sterile neutrinos.

4.3 The Chiral Mass Matrix

To find the physical mass eigenstates, we rewrite (10) using the basis of left-handed fields. We define the interaction basis spinor $\Psi = (\nu_L, \nu_R^c)^T$. Utilizing the Majorana identity, the Lagrangian allows the explicit matrix expansion:

$$\mathcal{L}_{\text{Mass}} = \frac{1}{2} \bar{\Psi}^c \mathcal{M} \Psi + \text{h.c.},$$

where the mass matrix $\mathcal{M}$ incorporates the chiral modifications from the CDE defined in (9):

$$\mathcal{M} = \begin{pmatrix} M_L & M_D \\ M_D^T & M_R \end{pmatrix} = \begin{pmatrix} M_L & m e^{-2i\alpha} \\ m e^{-2i\alpha} & M_R \end{pmatrix}. \quad (11)$$

4.4 The Neutrino Chiral Mass

To account for the smallness of neutrino masses, we adopt the following identifications in the matrix (11):

1. M_R : Heavy sterile scale ($M_R \gg m$).
2. M_L : Assumed $M_L \approx 0$.
3. M_D : Given by $me^{-2i\alpha}$.

Based on these assumptions, the mass matrix is given by:

$$\mathcal{M} \approx \begin{pmatrix} 0 & me^{-2i\alpha} \\ me^{-2i\alpha} & M_R \end{pmatrix}. \quad (12)$$

Finding the eigenvalues of (12) and applying the seesaw approximation ($m \ll M_R$), we obtain two mass terms:

1. **Heavy Mass** (m'_1): Sterile Dominant State.

$$m'_1 \approx M_R + \frac{m^2 e^{-4i\alpha}}{M_R} \approx M_R \quad (13)$$

2. **Light Mass** (m'_2): Active Dominant State.

$$m'_2 \approx -\frac{(me^{-2i\alpha})^2}{M_R} = -\frac{m^2}{M_R} e^{-4i\alpha} \quad (14)$$

4.5 Mass Suppression and CP Violation

The application of the Chiral Seesaw Mechanism yields the following expression for the active neutrino mass based on (14):

$$m'_\nu \approx \frac{m^2}{M_R} e^{-4i\alpha}. \quad (15)$$

This equation implies two significant physical results:

4.5.1 Mass Suppression

The magnitude of the active neutrino mass is given by $|m'_\nu| \approx m^2/M_R$. This confirms the standard seesaw suppression mechanism. For instance, if we assume $m \sim 100$ GeV and $|m'_\nu| \sim 0.1$ eV, the mechanism necessitates a heavy scale of $M_R \sim 10^{14}$ GeV.

4.5.2 The Chiral Phase and CP Violation

A novel feature of this model is that the **phase** of the light neutrino mass in (15) is directly determined by the chiral angle α :

$$\text{Phase}(m'_\nu) = e^{-4i\alpha}. \quad (16)$$

This dependence has profound implications:

- **Leptonic CP Violation:** In a three-generation scenario, this complex phase translates into Majorana phases in the PMNS mixing matrix. A value of $\alpha \neq k\pi/4$ implies a novel source of **leptonic CP violation**.

- **BSM Link:** This establishes a direct link between the theoretical angle α of the CDE and a physical observable mediated by the seesaw scale M_R .

5 Higger spin

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6 Results

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7 Conclusions

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References

- [1] Watson, T.B., Musielak, Z.E.: Chiral symmetry in Dirac equation and its effects on neutrino masses and dark matter. *International Journal of Modern Physics A* **35**(30), 2050189 (2020) <https://doi.org/10.1142/S0217751X20501894>
- [2] Watson, T., Musielak, Z.: Chiral Dirac Equation and its spacetime and CPT Symmetries. *Symmetry* **13**(9), 1608 (2021) <https://doi.org/10.3390/sym13091608>
- [3] Watson, T.B.: Chirality in quantum field theory and its role in the standard model and beyond. PhD thesis, University of Texas at Arlington (2022)
- [4] Bargmann, V., Wigner, E.P.: Group Theoretical Discussion of Relativistic Wave Equations. In: Wightman, A.S. (ed.) *Part I: Particles and Fields. Part II: Foundations of Quantum Mechanics*, pp. 82–94. Springer, Berlin, Heidelberg (1997). https://doi.org/10.1007/978-3-662-09203-3_6
- [5] Peskin, M.E., Schroeder, D.V.: *An Introduction to Quantum Field Theory*. The Advanced Book Program. CRC Press, Taylor & Francis Group, Boca Raton, London, New York (2019)
- [6] Steane, A.M.: An Introduction to Spinors. arXiv (2013). <https://doi.org/10.48550/ARXIV.1312.3824>
- [7] Wigner, E.: On unitary representations of the inhomogeneous lorentz group. *Annals of Mathematics* **40**(1), 149 (1939) <https://doi.org/10.2307/1968551>